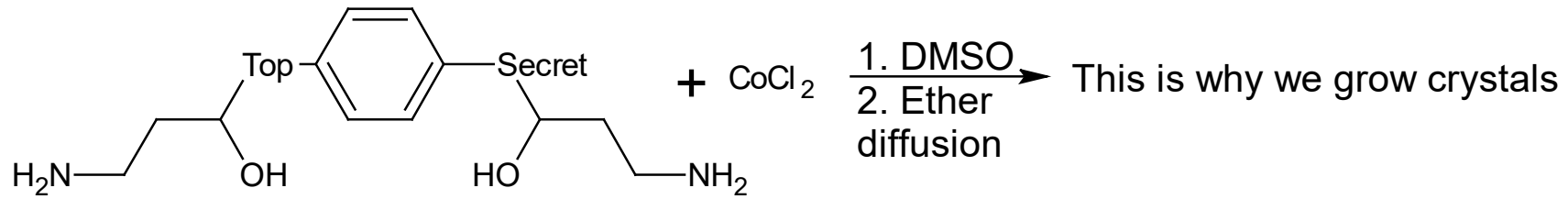
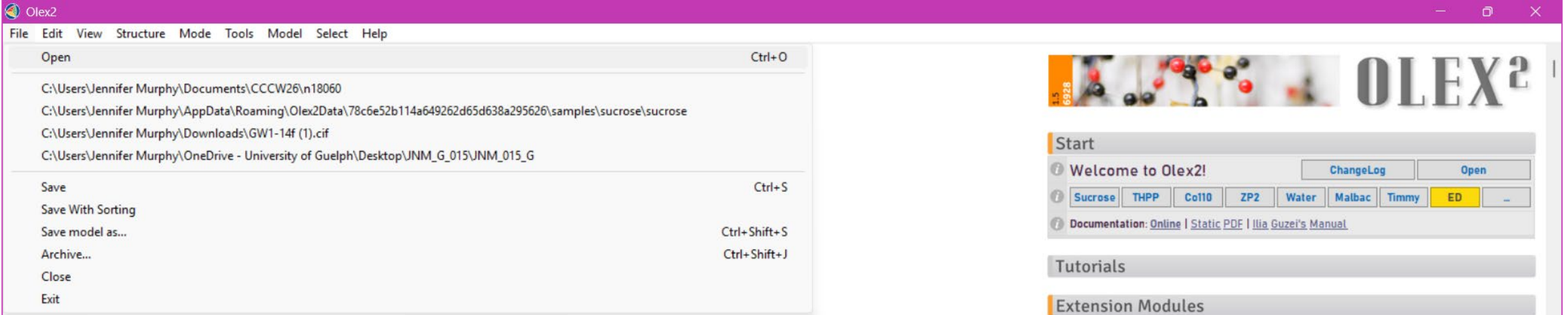


Olex2 – Getting Started

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Need a file with basic information and a reflection file .hkl

Navigate to your data folder and select your .ins (sometimes .p4p)

Careful about putting your data in the subfolder of a subfolder with a long pathname

Hit the “work” tab



OLEX²

Home Work View Tools Info

Start

Welcome to Olex2! [ChangeLog](#) [Open](#)

[Sucrose](#) [THPP](#) [Co110](#) [ZP2](#) [Water](#) [Malbac](#) [Timmy](#) [ED](#) [...](#)

Documentation: [Online](#) | [Static PDF](#) | [Ilia Guzei's Manual](#)

Tutorials

Extension Modules

Various extension modules for Olex2 are currently under development. If you are interested in testing these, please sign up below! Please **do not use temporary emails** - we may need to contact you in the case of new releases or feature updates.

e-mail Reference

Modules [Please Choose](#)

Select a module from the drop-down menu or type the name of a custom module

Install offline module in ZIP format. [Install Offline](#)

Settings

News

n18060 PI
C:\Users\Jennifer Murphy\Documents\CCCW26\n18060.ins

C12H42Cl4Co2O6S6

a = 8.9018(2) α = 81.945(2)° Z = 2
b = 9.4275(2) β = 81.395(2)° Z' = 1
c = 18.2484(4) γ = 89.429(1)° V = 1499.16(6)

5079.0
5079
0

Solution

d min (CuKα)	0.84	I/σ(I)	32.9	Rint	7.24%	Full 132.7°	96.4
2θ=132.7°		m=9.16					
Shift	n/a	Max Peak	n/a	Min Peak	n/a	Goof	n/a

Structure n18060 loaded

Home **Work** View Tools Info

Solve  Refine  Draw  Report 

Hit this dropdown arrow, **NOT** the word “Solve”!

Toolbox Work

Labels ☐ Label H ☐ No symm

...  

     Z=

MAP Diff

Modelling and Disorder Tools

Peak & Uiso Sliders

Growing

Finishing

History

Select

Naming

Sorting

Select the program that you want to use to solve your structure. ShelxT has become very popular, and we will use it today, but there are other powerful ways to solve structures, especially if you are encountering problems.

Sometimes you may want to look at other possible spacegroups (you won't have Rigaku's Explain as a possibility; Bruker users could also do this in XPREP)

n18060 P1
 C:\Users\Jennifer Murphy\Documents\CCCW26\n18060.ins
C12H42Cl4Co2O6S6

a = 8.9018(2) α = 81.945(2)° Z = 2
 b = 9.4275(2) β = 81.395(2)° Z' = 1
 c = 18.2484(4) γ = 89.429(1)° V = 1499.16(6)

d min (CuKα) 0.84 I/σ(I) 32.9 Rint 0.0916 7.24% Full 132.7° 96.4
 2θ=132.7°

Shift n/a Max Peak n/a Min Peak n/a GooF n/a

Structure n18060 loaded

Home **Work** **View** **Tools** **Info**

Solve **Refine** **Draw** **Report**

	R1	Rweak	Alpha S	sAbs Orientation	n Space	group F
ShelXT	0.103	0.016	0.051 0	00 as input	P-1	---
	0.219	0.021	0.000 0	00 as input	P1	---

Program: ShelXT (selected) Method: Intrinsic Phasing
 Reflections: Thu May 14 13:28:02 2026
 Composition: olex2.solve 25.0 Å³ Z = 2 Z' = 1
 Space Group: [Suggest SG](#) P-1
 Solution Settings Extra

Toolbox Work

Labels: ☐ Label H ☐ No symm Hide
 C H Cl Co O S ... Add H

Once you have selected your method for solving the structure, hit **"Solve"**

This is what you will see if you hit ‘suggest SG’

3. Cell centering test

Centering	Strong	I/Count	Count	Weak	I/Count	Count
P	-		0	-		0
A	316.75(0.30)		3977	317.70(0.29)		3982
B	315.31(0.28)		3971	319.14(0.31)		3988
C	308.11(0.29)		3989	326.38(0.30)		3970
F	305.70(0.39)		1989	321.07(0.25)		5970
I	317.66(0.30)		3986	316.79(0.29)		3973
R	321.59(0.36)		2660	315.03(0.26)		5299

Chosen lattice(s): P

Current space group: P-1 (P -1) #2

Possible space groups:

Noncentrosymmetric:

P1 (#1, Laue class -1, Point group 1)

Centrosymmetric:

P-1 (#2, Laue class -1, Point group -1)

>>|

n18060 PI
C:\Users\Jennifer Murphy\Documents\CCCW26\n18060.ins

C12H42Cl4Co2O6S6

a = 8.9018(2) α = 81.945(2)° Z = 2
b = 9.4275(2) β = 81.395(2)° Z' = 1
c = 18.2484(4) γ = 89.429(1)° V = 1499.16(6)

Solution
d min (CuKα) 0.84 I/σ(I) 32.9 Rint m=9.16 7.24% Full 132.7° 96.4
Shift n/a Max Peak n/a Min Peak n/a GooF n/a

Structure n18060 loaded

Home Work View Tools Info
Solve Refine Draw Report

ShelXT

R1	Rweak	Alpha S	sAbs Orientation	n Space	group F
0.103	0.016	0.051 0	00 as input	P-1	---
0.219	0.021	0.000 0	00 as input	P1	---

Program ShelXT **Method** Intrinsic Phasing

Reflections n18060.hkl Thu May 14 13:28:02 2026

Composition C12 H42 Cl4 Co2 O6 S6 25.0 Å³ Z = 2 Z' = 1

Space Group **Suggest SG** P-1

Solution Settings Extra

Toolbox Work

Labels [] Label H No symm Hide

C H Cl Co O S ... Add H

Q to QH Hx Q H Z' = 1 OK

Fix all AFIX Free XYZ Free ADP Clear HFIX Free H ADP

MAP Diff Show Map Map Settings

Modelling and Disorder Tools

Peak & Uiso Sliders

Growing

Finishing

History

Select

 C
 Cl
 Co
 O
 S

ShelXT Results

Structure s

1 Centrosymmetric and 1 non-centrosymmetric space groups evaluated

Space group determination: 0.004 secs

R1	Rweak	Alpha	SysAbs	Orientation	Space group	Flack_x	File	Formula
0.103	0.016	0.051	0.00	as input			n18060_a	C13 C19 Co2 O6 S
0.219	0.021	0.000	0.00	as input		0.49	n18060_b	C29 C15 O12 S4

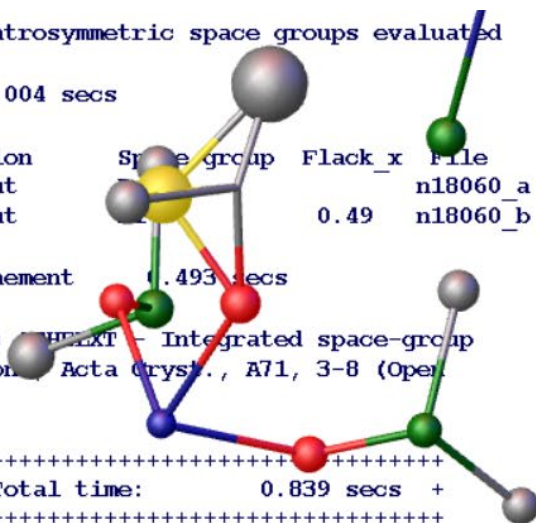
Assign elements and isotropic refinement 0.493 secs

Please cite: G.M. Sheldrick (2015) SHELXT – Integrated space-group and crystal-structure determination. *Acta Cryst.*, A71, 3-8 (Open Access) if SHELXT proves useful.

```

+++++
+  SHELXT finished at 14:11:59    Total time:      0.839 secs  +
+++++

```



n18060
C:\Users\Jennifer.Murphy\Documents\CCCW26\n18060.res

C₁₂H₄₂Cl₄Co₂O₆S₆

a = 8.9018(2) α = 81.945(2)° Z = 2
b = 9.4275(2) β = 81.395(2)° Z' = 1
c = 18.2484(4) γ = 89.429(1)° V = 1499.16(6)

Solution

d min (CuKα) 2θ = 132.7° 0.84 I/σ(I) 32.9 Rint m = 9.16 7.24% Full 132.7° 96.4%
Shift n/a Max Peak n/a Min Peak n/a GooF n/a

Running ShelXT

Home **Work** **View** **Tools** **Info**

Solve **Refine** **Draw** **Report**

ShelXT

R1	Rwack	Alpha S	sAbs Orientation	n Space	group F
0.103	0.016	0.051 0	00 as input	P-1	---
0.219	0.021	0.000 0	00 as input	P1	---

Program ShelXT **Method** Intrinsic Phasing

Reflections n18060.hkl Thu May 14 14:11:59 2026

Composition C12 H42 Cl4 Co2 O6 S6 25.0 Å³ Z = 2 Z' = 1

Space Group [Suggest SG](#) P-1

Solution Settings Extra

Toolbox Work

Labels ☐ Label H ☐ No symm [Hide](#)

C **H** **Cl** **Co** **O** **S** ... [Add H](#)

Q to C **Q to H** **Z' = 1** [OK](#)

Fix all **AFIX** **Free XYZ** **Free ADP** **Clear HFIX** **Free H ADP**

MAP Diff [Show Map](#) [Map Settings](#)

Modelling and Disorder Tools

Peak & Uiso Sliders

Growing

Finishing

History

Select

Does this solution make sense? Go back to your reaction.

A few things to try:

F_n F₄

Ctrl T

Type “grow”

Type “fuse”

Go to “Refine” arrow
(**Don't** hit “Refine” yet!)

The screenshot displays the Olex2 software interface for file n18060.res. The top section shows the chemical formula $C_{12}H_{42}Cl_4Co_2O_6S_6$ and unit cell parameters: $a = 8.9018(2)$, $b = 9.4275(2)$, $c = 18.2484(4)$; $\alpha = 81.945(2)^\circ$, $\beta = 81.395(2)^\circ$, $\gamma = 89.429(1)^\circ$; $Z = 2$, $Z' = 1$, $V = 1499.16(6)$. It also lists $R_1 = 11.72\%$ and $wR_2 = 33.96\%$. A table of data follows:

d min (CuK α)	0.84	I/o(I)	32.9	Rint	7.24%	Full 132.7 $^\circ$	96.4
2 θ =132.7 $^\circ$				m=9.16			
Shift	-15.000	Max Peak	10.3	Min Peak	-2.7	Goof	2.835

 Below this, a 'Refinement finished' message is shown. The main menu includes Home, Work, View, Tools, and Info. The 'Solve' menu is open, showing options: Solve, Refine (highlighted with a red arrow), Draw, Report. The 'Refine' option is selected, and the 'Program' is set to ShelXL. The 'hkl file' is n18060.hkl. The 'Weight' is set to .100 | .112. The 'Use Mask' checkbox is checked. The 'Refinement Settings Extra' section is visible. The 'Toolbox Work' section shows various tools like Labels, C H Cl Co O S, and buttons for Fix all, AFIX, Free XYZ, Free ADP, Clear HFIX, and Free H ADP. The 'MAP' section is set to Diff. The 'Modelling and Disorder Tools' section is also visible.

Disagree with the preliminary atom assignments?

Click on atoms you would like to reassign, and click on the identity that you think they should be.

(The “...” option will allow you to select atom types that were not included in your original formula.)

Do you see atoms that shouldn't be there? Click on them and hit “**Delete**” on your keyboard.

Note: Sometimes it can be tricky to select atoms (like in this case.) You can also select bonds and delete those (this makes selecting the atom easier.) You can also hold down Shift and your left mouse button to draw a box around the atoms you want to select.

n18060 PI
 C:\Users\Jennifer Murphy\Documents\CCCW26\n18060.res

C₁₂H₄₂Cl₄Co₂O₆S₆

a = 8.9018(2)	α = 81.945(2)°	Z = 2	42.7	R₁ 11.72 %
b = 9.4275(2)	β = 81.395(2)°	Z' = 1	5079	
c = 18.2484(4)	γ = 89.429(1)°	V = 1499.16(6)	119	wR₂ 33.96 %

d min (CuKα) 2θ = 132.7°	0.84	I/σ(I)	32.9	Rint m = 9.16	7.24%	Full 132.7°	96.4
Shift	-15.000	Max Peak	10.3	Min Peak	-2.7	Goof	2.835

Refinement finished

Home Work View Tools Info

Solve Refine Draw Report

Program: ShelXL, LS, Client, Cycles: 10, Peaks: 59

hkl file: n18060.hkl, Thu May 14 14:43:48 2026

Weight: .100 | .112, .000 | 49.479, EXTI, SWAT, No ACTA

Use Mask: Use a solvent mask (smtbx.mask or SQUEEZE)

Refinement Settings Extra

Toolbox Work

Labels: [] Label H No symm Hide

C H Cl Co O S ... Add H

QC QH Hx Q H OK

Fix all AFIX Free XYZ Free ADP Clear HFIX Free H ADP

MAP Diff Show Map Map Settings

Modelling and Disorder Tools

Peak & Uiso Sliders

Growing

IN OLEX2 THERE ARE MANY, MANY WAYS TO ACHIEVE THE SAME GOAL! SHARE YOUR TIPS! :-D

Select your refinement program (**ShelXL**)

Note that it has selected the reflection file with the same file name; if you have other reflection files in this folder, you can select the one you want to use.

If you feel like you are in a good place to get started with your refinement, hit “**Refine**”

n18060 PI
C:\Users\Jennifer Murphy\Documents\CCCW26\n18060.res

C12H42Cl4Co2O6S6

a = 8.9018(2)	α = 81.945(2)°	Z = 2	42.7	R_1 11.72 %
b = 9.4275(2)	β = 81.395(2)°	Z' = 1	5079	
c = 18.2484(4)	γ = 89.429(1)°	V = 1499.16(6)	119	wR_2 33.96 %

d min (CuK α)	0.84	I/o(I)	32.9	Rint	7.24%	Full 132.7°	96.4
2 θ =132.7°				m=9.16			
Shift	-15.000	Max Peak	10.3	Min Peak	-2.7	Goof	2.835

Refinement finished

Home **Work** View Tools Info

Solve **Refine** Draw Report

Program: ShelXL LS. Client: Cycles: 10 Peaks: 59

hkl file: n18060.hkl Thu May 14 14:43:48 2026

Weight: .100 | .112 .000 | 49.479 EXTI SWAT No ACTA

☐ Use Mask Use a solvent mask (smtbx.mask or SQUEEZE)

Refinement Settings Extra

Toolbox Work

Labels: Label H No symm Hide

C H Cl Co O S ... Add H

QC QH Hx QH H OK

Fix all AFIX Free XYZ Free ADP Clear HFIX Free H ADP

MAP Diff Show Map Map Settings

Modelling and Disorder Tools

Peak & Uiso Sliders

How are things looking?

Ctrl T

Ctrl Q

Hover over a Q peak.

Scroll down on your mouse and the smallest peaks will start to be hidden.

If you would like to find your largest Q peak without hiding the others:

1. Intensity of the Q peak colour corresponds to size
2. Hitting this will highlight the maximum peak. What's up with that largest peak?
3. "Fn F3" will label everything. Those labels are terrible.

n18060 PI
 C:\Users\Jennifer Murphy\Documents\CCCW26\n18060.res

C12H42Cl4Co2O6S6

$a = 8.9018(2)$ $\alpha = 81.945(2)^\circ$ $Z = 2$ $R_1 = 11.72\%$
 $b = 9.4275(2)$ $\beta = 81.395(2)^\circ$ $Z' = 1$ $wR_2 = 33.96\%$
 $c = 18.2484(4)$ $\gamma = 89.429(1)^\circ$ $V = 1499.16(6)$

$d_{min} (CuK\alpha) = 0.84$ $I/\sigma(I) = 32.9$ $R_{int} = 7.24\%$ $Full 132.7^\circ = 96.4\%$
 $2\theta = 132.7^\circ$ $Shift = -15.000$ $Max Peak = 10.3$ $Min Peak = -2.7$ $Goof = 2.835$

Refinement finished

Home **Work** **View** **Tools** **Info**
Solve **Refine** **Draw** **Report**

Program: ShelXL LS. Client Cycles: 10 Peaks: 59
 hkl file: n18060.hkl Thu May 14 14:43:48 2026
 Weight: ☐ .100 | .112 .000 | 49.479 EXTI ☐ SWAT ☐ No ACTA
☐ Use Mask Use a solvent mask (smtbx.mask or SQUEEZE)
 Refinement Settings Extra

Toolbox Work

Labels: ☐ Label H ☐ No symm Hide
 C H Cl Co O S ... Add H
 QC to QH Hx QH H OK
 Fix all AFIX Free XYZ Free ADP Clear HFIX Free H ADP
 MAP: Diff Show Map Map Settings
 Modelling and Disorder Tools
 Peak & Uiso Sliders
 Growing
 Finishing

History

Select

Naming

When you feel confident that all non-hydrogen atoms are in your model, but before you attempt to treat any disorder, you should give your atoms sensible names.

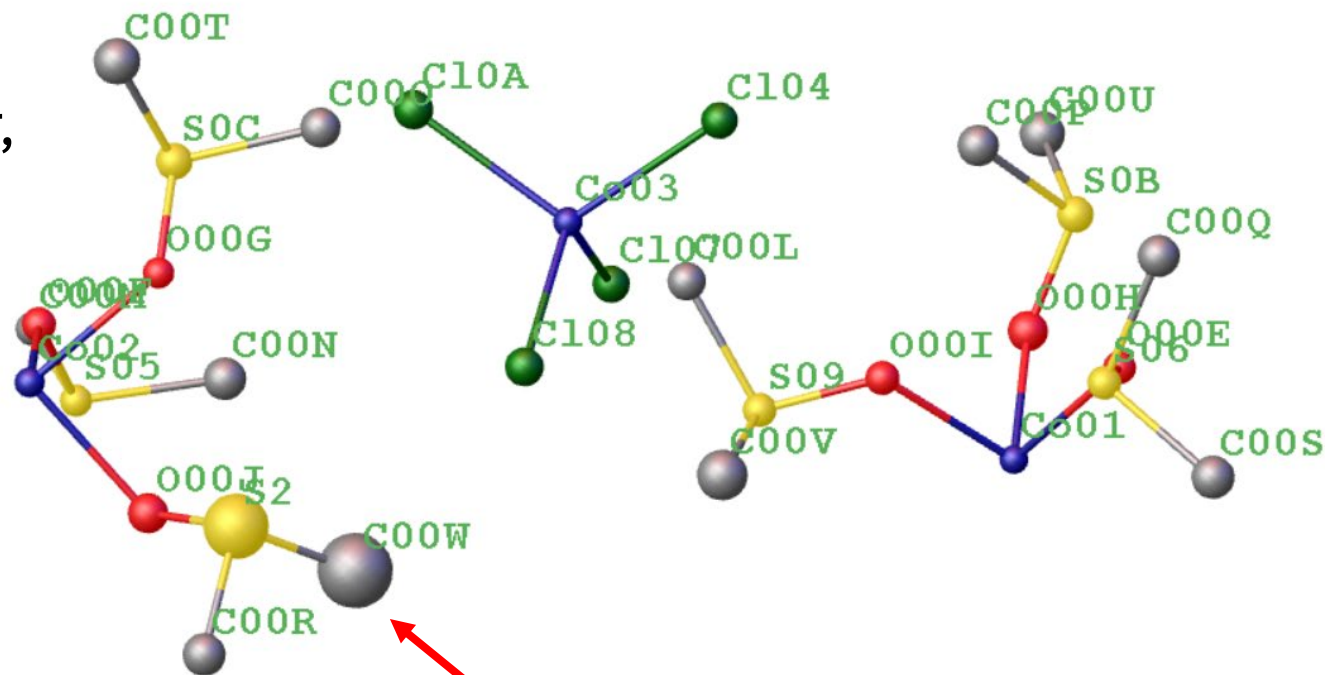
“**Ctrl Q**” to hide your Q peaks

Click on your cobalt atoms in the order that you would like to number them

At the prompt, type “**Name 1**”

Repeat for all atom types

Note: You don't have to start with 1 as your first number. For example, if you want to switch just part of a numbering sequence, select the atoms that are involved, and the number you want to start with.



If I suspect an atom, or group, to be involved in disorder, and if it makes sense to do so, I leave it for the last number.



Names okay? “**Refine**”

So far we’ve been refining everything isotropically, but we’ll want to switch to anisotropic refinement.

You can select **ACTA** to create a .cif

Then, hit the anisotropic ellipsoid to change how each atom is refined.

Everything okay so far? Hit “**Refine**”

+++++

Atom order after sorting:

```
Co1 S1 S2 S3 O1 O2 O3 C1 C2 C3 C4 C5 C6 Co2 S4 S5 S6 O4 O5 O6 C7 C8 C9 C10 C11
C12 Co3 C10A C104 C107 C108 Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q8 Q9 Q10 Q11 Q12 Q13 Q14 Q15
Q16 Q17 Q18 Q19 Q20 Q21 Q22 Q23 Q24 Q25 Q26 Q27 Q28 Q29 Q30 Q31 Q32 Q33 Q34 Q35
Q36 Q37 Q38 Q39 Q40 Q41 Q42 Q43 Q44 Q45 Q46 Q47 Q48 Q49 Q50 Q51 Q52 Q53 Q54 Q55
Q56 Q57 Q58 Q59
```

The screenshot shows the Olex2 software interface. At the top, the title bar reads "n18060" and the file path is "C:\Users\Jennifer Murphy\Documents\CCCW26\n18060.res". Below this, the chemical formula $C_{12}H_{42}Cl_4Co_2O_6S_6$ is displayed. The refinement statistics are shown in a table:

a = 8.9018(2)	$\alpha = 81.945(2)^\circ$	Z = 2	42.7	R_1	11.72 %
b = 9.4275(2)	$\beta = 81.395(2)^\circ$	Z' = 1	5079	wR_2	33.96 %
c = 18.2484(4)	$\gamma = 89.429(1)^\circ$	V = 1499.16(6)	119		

Below the table, the refinement statistics are summarized:

d min (CuK α)	0.84	I/o(I)	32.9	Rint	7.24%	Full 132.7 $^\circ$	96.4
Shift	-15.000	Max Peak	10.3	Min Peak	-2.7	Goof	2.835

The "Refinement finished" message is displayed. The "Work" tab is selected, and the "Refine" button is highlighted. A red arrow points from the "Refine" button to the "ACTA" option in the dropdown menu. The "ACTA" option is also highlighted in the dropdown menu. The "Toolbox Work" panel is visible at the bottom, showing various refinement tools and options.

Now how do things look?

How are your refinement values?

Where is that max peak?

Look at your Q peaks; do they make you think about hydrogen?

You can add hydrogen atoms into calculated positions by hitting the “**Add H**” button. “**Refine**”

If your data is good enough, and especially if you have hydrogen atoms involved in H-bonds, you will likely want to introduce them in their difference map positions and refine them (possibly with the use of restraints.)

The screenshot displays the Olex2 software interface. The top panel shows the file name 'n18060.res' and the chemical formula C12Cl4Co2O6S6. The refinement statistics are as follows:

Parameter	Value	Parameter	Value	Parameter	Value
a	8.9018(2)	α	81.945(2) $^\circ$	Z	2
b	9.4275(2)	β	81.395(2) $^\circ$	Z'	1
c	18.2484(4)	γ	89.429(1) $^\circ$	V	1499.16(6)

The R₁ value is 9.38% and the wR₂ value is 29.65%. The difference map statistics are:

Parameter	Value	Parameter	Value	Parameter	Value
d min (CuK α)	0.84	I/o(I)	32.9	Rint	7.24%
2 θ =132.7 $^\circ$				Full 132.7 $^\circ$	96.4
Shift	0.364	Max Peak	5.7	Min Peak	-4.8
				Goof	2.481

A warning message states: 'Warning: 2 atoms may be split and 0 atoms NPD'. The 'Refine' button is highlighted in the 'Solve' panel. The 'Toolbox Work' panel shows the 'Add H' button, which is highlighted with a red arrow. The 'Add H' button is located in the 'Labels' section of the 'Toolbox Work' panel.

1. Should all those hydrogens be there? Go back to your reaction conditions and think about charge balance.
2. Hydrogen atoms can “mask” disorder by mopping up electron density.

Go back and delete any hydrogen atoms that are relevant to these two points (click on them and hit “**Delete**” on your keyboard.)

“Refine”

My refinement values improved. Did yours? This wasn’t a sure thing since points 1 and 2 were likely to off-set each other (ie. Point 1 meant that there were H-atoms introduced where none existed but point 2 meant that some H-atoms might have been used to account for electron density associated with disorder.)

Before we start dealing with disorder, let’s do a few routine “things”.

At the prompt type “**edit lst**”

First, the atoms are not in a sensible order.

Close the .lst and sort your atoms

First, click on “**Sorting**” (if this menu isn’t open)

Next, decide on your sort order (I’ll just use the default) and hit “**Sort**”

“**Refine**”

You can also just type ‘**sort**’ in the command line

The screenshot shows the Olex2 software interface. The top bar displays the file name 'n18060.res' and the path 'C:\Users\Jennifer Murphy\Documents\CCCW26\n18060.res'. The main window shows the chemical formula $C_{12}H_{42}Cl_4Co_2O_6S_6$ and various refinement statistics. The 'Refinement finished' status is shown. The 'Work' tab is active, and the 'Sorting' menu is open. The 'Sort order' is set to 'Part', 'Z', 'Label', and 'None'. The 'Moiety' is set to 'From sort'. The 'Specific order' section has 'Atoms' and 'Moieties' fields. The 'Sort' button is highlighted with a red arrow.

a = 8.9018(2)		α = 81.945(2)°		Z = 2	42.7 5079 119	R ₁ wR ₂	11.72 % 33.96 %
b = 9.4275(2)		β = 81.395(2)°		Z' = 1			
c = 18.2484(4)		γ = 89.429(1)°		V = 1499.16(6)			

d min (CuKα)	2θ = 132.7°	0.84	I/o(I)	32.9	Rint m = 9.16	7.24%	Full 132.7°	96.4
Shift	-15.000	Max Peak	10.3	Min Peak	-2.7	Goof	2.835	

Refinement finished

Home Work View Tools Info

Solve Refine Draw Report

Program: ShelXL L.S. Client Cycles: 10 Peaks: 59

hkl file: n18060.hkl Thu May 14 14:43:48 2026

Weight: .100 | .112 .000 | 49.479 EXTI SWAT ACTA

Use Mask: Use a solvent mask (smtbx.mask or SQUEEZE)

Refinement Settings Extra

Toolbox Work

History

Select

Naming

Sorting

Sort order: Part Z Label None

Moiety: From sort Treat H atoms independently

Specific order: Atoms Moieties Reorder Move to first

Sort

Are you up-dating your weighting? (I've been waiting until I deal with my disorder, because I know I don't yet have my best model.

What will “Info” reveal?

n18060
C:\Users\Jennifer Murphy\Documents\CCCW26\n18060.res

C12H36Cl4Co2O6S6

a = 8.9018(2) α = 81.945(2)* Z = 2
b = 9.4275(2) β = 81.395(2)* Z' = 1
c = 18.2484(4) γ = 89.429(1)* V = 1499.16(6)

17.8
5079
286

R_1 **8.45 %**
 wR_2 **27.60 %**

d min (CuKα) 2θ = 132.7°	0.84	I/σ(I)	32.9	Rint m = 9.16	7.24%	Full 132.7°	96.4
Shift	0.000	Max Peak	5.0	Min Peak	-5.0	GooF	2.300

Warning: 2 atoms may be split and 0 atoms NPD

Home Work View Tools Info

n18060 PI
C:\Users\Jennifer Murphy\Documents\CCCW26\n18060.res

C12H36Cl4Co2O6S6

a = 8.9018(2) α = 81.945(2)* Z = 2
b = 9.4275(2) β = 81.395(2)* Z' = 1
c = 18.2484(4) γ = 89.429(1)* V = 1499.16(6)

17.8
5079
286

R_1 **8.45 %**
 wR_2 **27.60 %**

d min (CuKα) 2θ = 132.7°	0.84	I/σ(I)	32.9	Rint m = 9.16	7.24%	Full 132.7°	96.4
Shift	0.000	Max Peak	5.0	Min Peak	-5.0	GooF	2.300

Warning: 2 atoms may be split and 0 atoms NPD

Home Work View Tools Info

Solve Refine Draw Report

Program: ShelXL L.S. Client Cycles: 10 Peaks: 23

File: n18060.hkl Thu May 14 14:43:48 2026

Weight: ☐ .100 | .046 .000 | 36.975 EXTI ☐ SWAT ☒ ACTA

☐ Use Mask Use a solvent mask (smtbx.mask or SQUEEZE)

Refinement Settings Extra

Toolbox Work

History

Select

Naming

Sorting

Sort order: Part Z Label None

Moiety: From sort Treat H atoms independently ☐

Specific order

Atoms: Reorder ☐ Move to first ☐

Moieties: Sort

More info in our .lst!

Difference Map Peaks

Refinement Indicators

Bad Reflections

Reflection Statistics

Reflection Statistics Summary

At the prompt type **“edit lst”**

In the text file, search for

“disagreeable”

“split”

“Highest peak”

There is so much great information in this file!

Okay, at this point, we can either model this disorder, or leave that to someone else...

What time is it?

In your main screen “Ctrl T” until your have a blank background and “Ctrl Q” to show your Q-peaks, but scroll down on your mouse so that only the two highest ones are displayed.

Open Toolbox Work by clicking on it

There are multiple ways to treat disorder in OLEX2; you can click on the “i” to get info about the various settings that you can apply.

```
+++++
+ n18060      finished at 09:11:19   Total elapsed time:    0.25 secs +
+++++
```

Note: Q-peaks are invisible
The following atom(s) may be split:
S6 C12

>>|

The screenshot displays the Olex2 software interface. The top section shows the file name 'n18060' and the path 'C:\Users\Jennifer Murphy\Documents\CCCW26\n18060.res'. Below this, the chemical formula $C_{12}H_{36}Cl_4Co_2O_6S_6$ is shown. A red box highlights a warning icon. The main data table provides crystallographic parameters:

a = 8.9018(2)	α = 81.945(2)°	Z = 2	17.8	R_1	8.50 %
b = 9.4275(2)	β = 81.395(2)°	Z' = 1	5079	wR_2	27.69 %
c = 18.2484(4)	γ = 89.429(1)°	V = 1499.16(6)	285		

Below the table, a red box highlights a warning message: "Cell contents from UNIT instruction and atom list do not agree | Warning: 2 atoms may be split and 0 atoms NPD".

The interface includes a menu bar (File, Edit, View, Structure, Mode, Tools, Model, Select, Help) and a toolbar with icons for Solve, Refine, Draw, and Report. The 'Toolbox Work' panel is open, showing various settings for Labels, Modelling and Disorder Tools, and Peak & Uiso Sliders.

I recommend deleting the H-atoms on the C that is involved in the disorder.

I also like to make these atoms isotropic first.

Select the C and S that are involved in disorder, and hit “**Split**”, leaving the other settings as shown.

Hover your mouse over the group you want to split, and hold “**Ctrl**” and your left mouse button while moving your mouse around until the second component fits the two highest Q peaks

“**Refine**”

n18060 PI
C:\Users\Jennifer Murphy\Documents\CCCW26\n18060.res

C12H36Cl4Co2O6S6

a = 8.9018(2) α = 81.945(2)° Z = 2 17.8
b = 9.4275(2) β = 81.395(2)° Z' = 1 5079
c = 18.2484(4) γ = 89.429(1)° V = 1499.16(6) 285

R_1 **8.50 %**
 wR_2 **27.69 %**

d min (CuKα) 2θ = 132.7°	0.84	I/σ(I)	32.9	Rint m = 9.16	7.24%	Full 132.7°	96.4
Shift	0.010	Max Peak	5.1	Min Peak	-5.0	Goof	2.307

Cell contents from UNIT instruction and atom list do not agree | Warning: 2 atoms may be split and 0 atoms NPD

Home **Work** **View** **Tools** **Info**

Solve **Refine** **Draw** **Report**

Program: ShelXL L.S. Client Cycles: 10 Peaks: 26

hkl file: n18060.hkl Thu May 14 14:43:48 2026

Weight: ☐ .100 | .049 .000 | **37.389** EXTI ☐ SWAT ☐ **ACTA**

☐ Use Mask Use a solvent mask (smtbx.mask or SQUEEZE)

Refinement Settings Extra

Toolbox Work

Labels: ☐ Label H ☐ No symm Hide

C H Cl Co O S ... Add H

Q to C Q to H H X H H H H Z' = 1 OK

Fix all AFX Free XYZ Free ADP Clear HFX Free H ADP

MAP: Diff Show Map Map Settings

Modelling and Disorder Tools

Fit | **Split** with SAME PART auto FVAR auto

FVAR: |

☐ Show/Hide the list of restraints No Restraints are shown 2.0

Peak & Uiso Sliders

Growing

Finishing

History

If you aren't happy about how these two atoms were split, you can always go back in history to an earlier model.

Just click on any of the Histogram bars, and the model that corresponded to that round of least squares will pop back up on your screen!

If you just don't like how you split your atoms, and want a second try at it, hit the last bar.

n18060 PI
C:\Users\Jennifer Murphy\Documents\CCCW26\n18060.res

C₁₂H₃₀Cl₄Co₂O₆S₆

a = 8.9018(2)	α = 81.945(2)°	Z = 2	17.9	R₁ 4.89 %
b = 9.4275(2)	β = 81.395(2)°	Z' = 1	5079	
c = 18.2484(4)	γ = 89.429(1)°	V = 1499.16(6)	283	wR₂ 15.34 %

d min (CuKα) 2θ = 132.7°	0.84	I/σ(I)	32.9	Rint m = 9.16	7.24%	Full 132.7°	96.4
Shift	-0.001	Max Peak	0.9	Min Peak	-1.1	Goof	1.250

Cell contents from UNIT instruction and atom list do not agree

Home **Work** **View** **Tools** **Info**

Solve **Refine** **Draw** **Report**

Program: ShelXL | L.S. | Client: ☐ Cycles: 10 | Peaks: 36

hkl file: n18060.hkl | Thu May 14 14:43:48 2026

Weight: ☐ .100 | .049 | .000 | 5.029 | EXTI ☐ SWAT ☐ ACTA ☒

☐ Use Mask Use a solvent mask (smtbx.mask or SQUEEZE)

Refinement Settings Extra

Toolbox Work

History

Scale: 5

[Reload ins](#) [Create Branch](#) [Create Snapshot](#)

History Tree

Select

Naming

Sorting

You should notice that after you split your atoms they now look like they are being refined isotropically (because they are; anisotropic refinement can also “mop up” electron density, so we refine disorder isotropically first.) Notice the nice work OLEX2 did naming the second disorder component?

You should also notice that the H-atoms on the “not disordered” carbon disappeared. Why do you think that is?

Peter Mueller would tell you that you will never refine disorder without restraints. Let’s open our .ins to see what was added.

“edit .ins”

```
TITL n18060_a.res in P-1
REM Old TITL n18060_a.res in P-1
REM SHELXT solution in P-1: R1 0.103, Rweak 0.016, Alpha 0.051
REM <I/s> 0.000 for 0 systematic absences, Orientation as input
REM Formula found by SHELXT: C13 C19 Co2 O6 S
CELL 1.54184 8.9018 9.4275 18.2484 81.945 81.395 89.429
ZERR 2 0.0002 0.0002 0.0004 0.002 0.002 0.001
LATT 1
SFAC C H Cl Co O S
UNIT 24 72 8 4 12 12
SADI S6 C11 S6A C11
SADI S6 C12 S6A C12A
SADI 0.04 06 C12 06 C12A
SADI 0.04 C11 C12 C11 C12A

L.S. 10
PLAN 24
CONF
LIST 6
fmap 2
MORE -1
ACTA
BOND $H
WGHT 0.1
FVAR 0.46967 0.6548
REM <olex2.extras>
REM <HklSrc "%.\\n18060.hkl">
REM </olex2.extras>
```

Peter says it here:

<https://www.tandfonline.com/doi/pdf/10.1080/08893110802547240?needAccess=true>

We can add additional restraints and constraints.

First, open up the “**Tools**” menu

Then, open up “**Shelx Compatible Restraints**” or “**Constraints**”

You should not simultaneously refine occupancy and displacements (at least on your first round of least squares).

We can apply an **EADP** constraint to force selected displacements to be identical.

Later, we can edit these constraints to replace with **RIGU** restraints.

Shelx instructions!

http://shelx.uni-goettingen.de/shelxl_html.php

c = 18.2484(4)		γ = 89.429(1)°		V = 1499.16(6)		308	wR_2	14.14 %
d min (CuKα)	0.84	I/σ(I)	32.9	Rint	m=9.16	7.24%	Full 152.7°	96.4
Shift	0.001	Max Peak	0.5	Min Peak	-1.1	Goof		1.154

Refinement finished

Home Work View **Tools** Info

FragmentDB

AC7

Twinning

Images

Maps

Chemical Tools

Olex2 Constraints Restraints

Shelx Compatible Constraints

Shelx Compatible Restraints

Hydrogen Atoms

Disorder

RIGU!

<http://journals.iucr.org/a/issues/2012/04/00/pc5011/pc5011.pdf>

If everything looked stable, hit your **anisotropic button** again, and “**Refine**”. If things still look good, you will want to add H-atoms.

This time, just select the carbon atoms in the disordered group (if you hit “**Add H**” before doing this, it will add back in the O-H hydrogen atoms that you already deleted.)

Notice how the carbon atom that was not split now has six hydrogens on it? Let’s see how this makes sense by selecting to view only one “**part**” at a time.

You can also select to update your weights now.

n18060 PI
 C:\Users\Jennifer Murphy\Documents\CCCW26\18060.res

C12H36Cl4Co2O6S6

a = 8.9018(2)	α = 81.945(2)°	Z = 2	16.6	R_1 4.54 %
b = 9.4275(2)	β = 81.395(2)°	Z' = 1	5079	
c = 18.2484(4)	γ = 89.429(1)°	V = 1499.16(6)	306	wR_2 14.14 %

d min (CuKα) 2θ = 132.7°	0.84	I/σ(I)	32.9	Rint m = 9.16	7.24%	Full 132.7°	96.4
Shift	0.001	Max Peak	0.5	Min Peak	-1.1	Goof	1.154

Refinement finished

Home **Work** **View** **Tools** **Info**

Solve **Refine** **Draw** **Report**

Program: ShelXL | LS. | Client | Cycles: 10 | Peaks: 24

hkl file: n18060.hkl | Thu May 14 14:43:48 2026

Weight: ☒ | .100 | .046 | .000 | 3.466 | EXTI ☐ SWAT ☐ ACTA

☐ Use Mask | Use a solvent mask (smtbx.mask or SQUEEZE)

Refinement Settings Extra

Toolbox Work

Labels: | ☐ Label H ☐ No symm | Hide

C H Cl Co O S ... | Add H

QC QH H H H H | Z' = 1 | OK

Fix all | AFIX | Free XYZ | Free ADP | Clear HFIX | Free H ADP

MAP: Diff | Show Map | Map Settings

Modelling and Disorder Tools

0 | 1 | 0 | 2 | All | Set ☒ Unique ☒ Labels |

Fit | Split with SAME | PART auto | FVAR auto

FVAR: | 21 0.655 |

☐ Show/Hide the list of restraints | No Restraints are shown | 2.0

Peak & Uiso Sliders

Growing

Finishing

In this tutorial I have deliberately not included screenshots of my model because I didn't want to give away the answer!

I've posted this as a file that you can edit. Why don't you go back, work through it again, and add your own notes and screenshots? You can start from scratch by going to Model → Reset

